Error

- ★ Data are transmitted in the network.
- ★ The data can be corrupted during transmission.
- ★ Transmission error.
- ★ For reliable communication, errors must be detected and corrected.
- ★ Error detection and correction are implemented either at the data link layer or the transport layer of the OSI model.

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Types of error

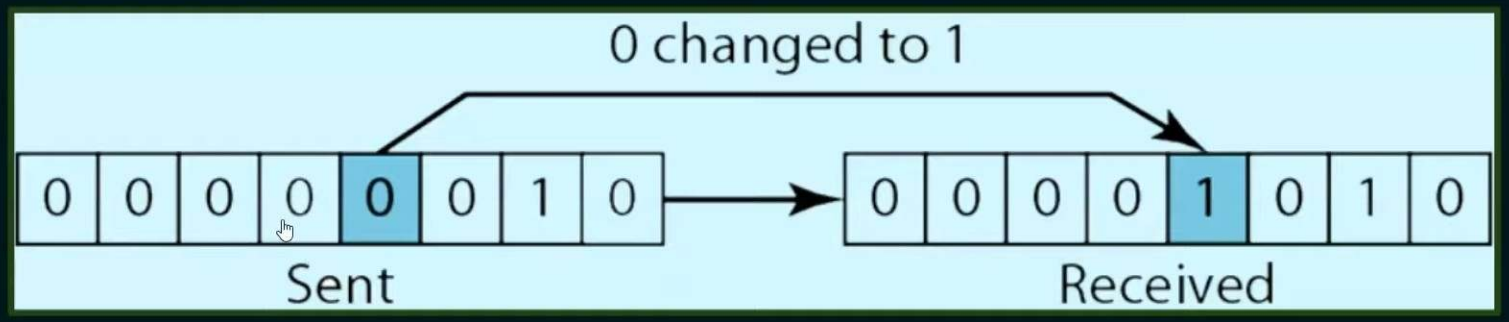
1. Bit Error.
2. Burst Error.

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Bit error

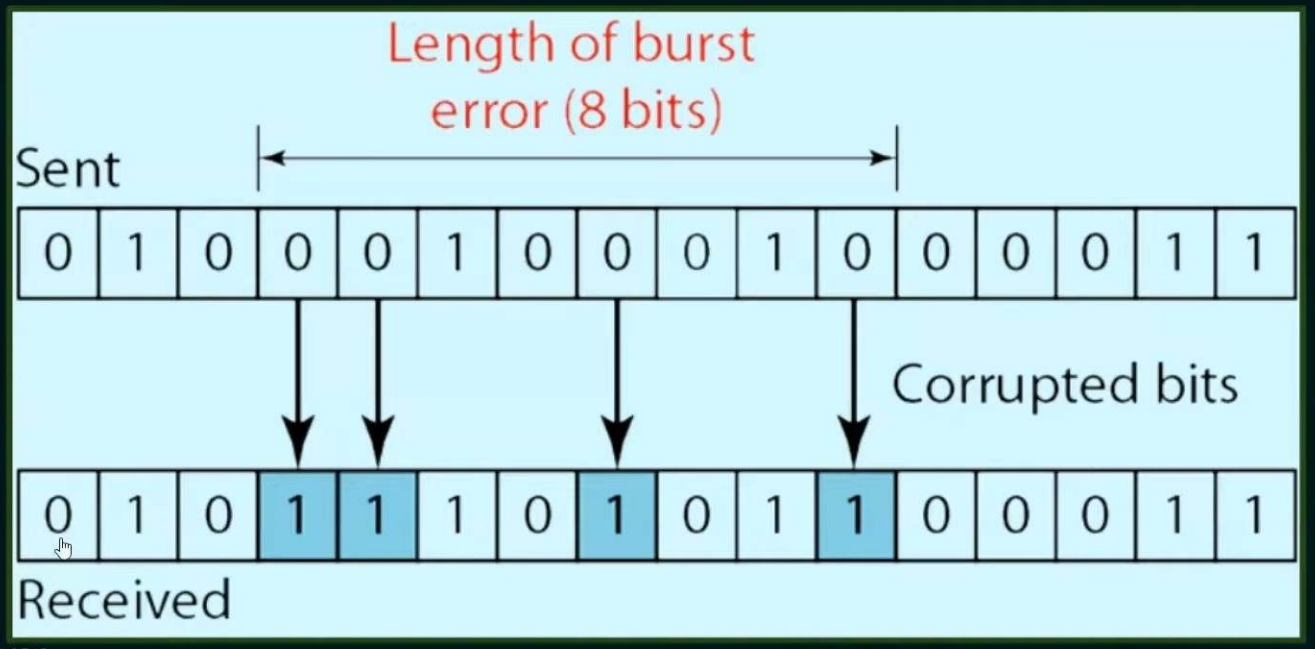
- ★ a.k.a single bit error.
- ★ In a single bit error, only 1 bit in the data unit has been changed.



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Burst error

★ In burst error, 2 or more bits in the data unit have changed.



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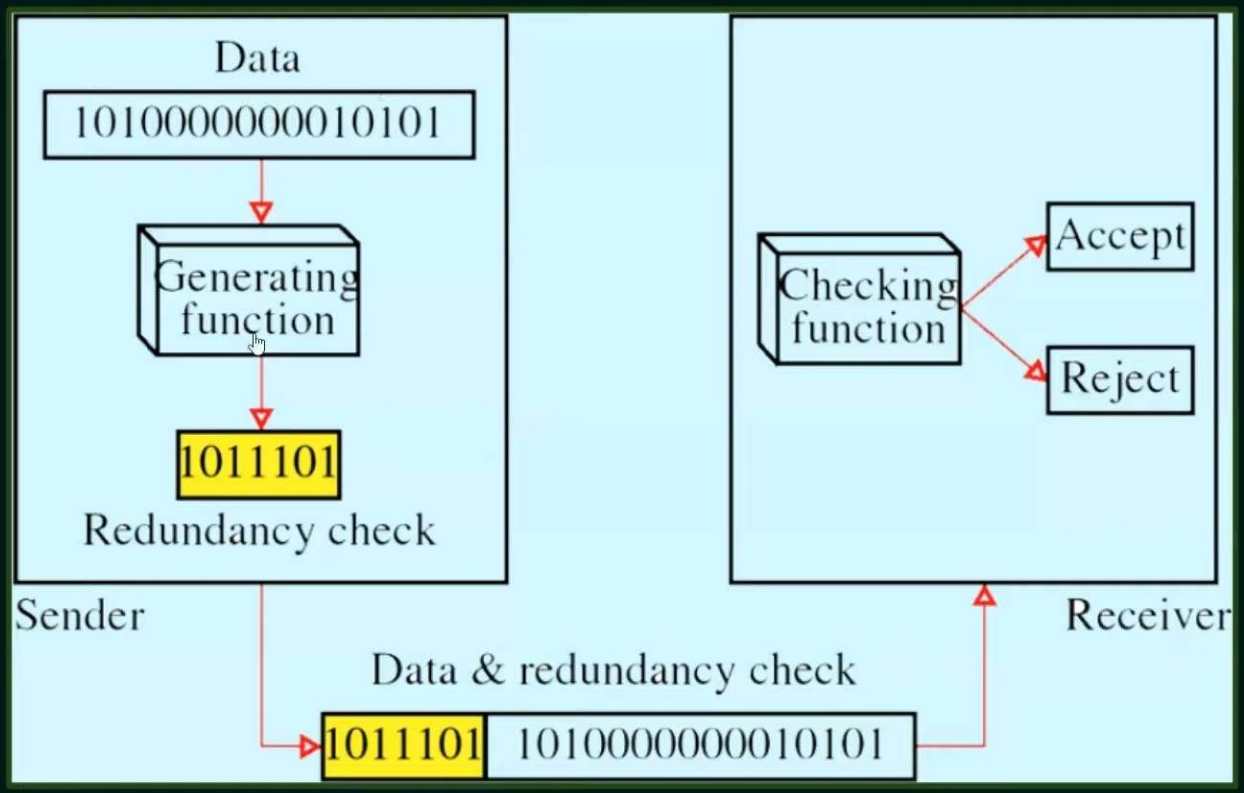
How to detect the errors?

- ★ Error detection means to decide whether the received data is correct or not without having a copy of the original message.
- ★ To detect or correct errors, we need to send some extra bits with the data.
- ★ The extra bits are called as redundant bits.

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REDUNDANCY



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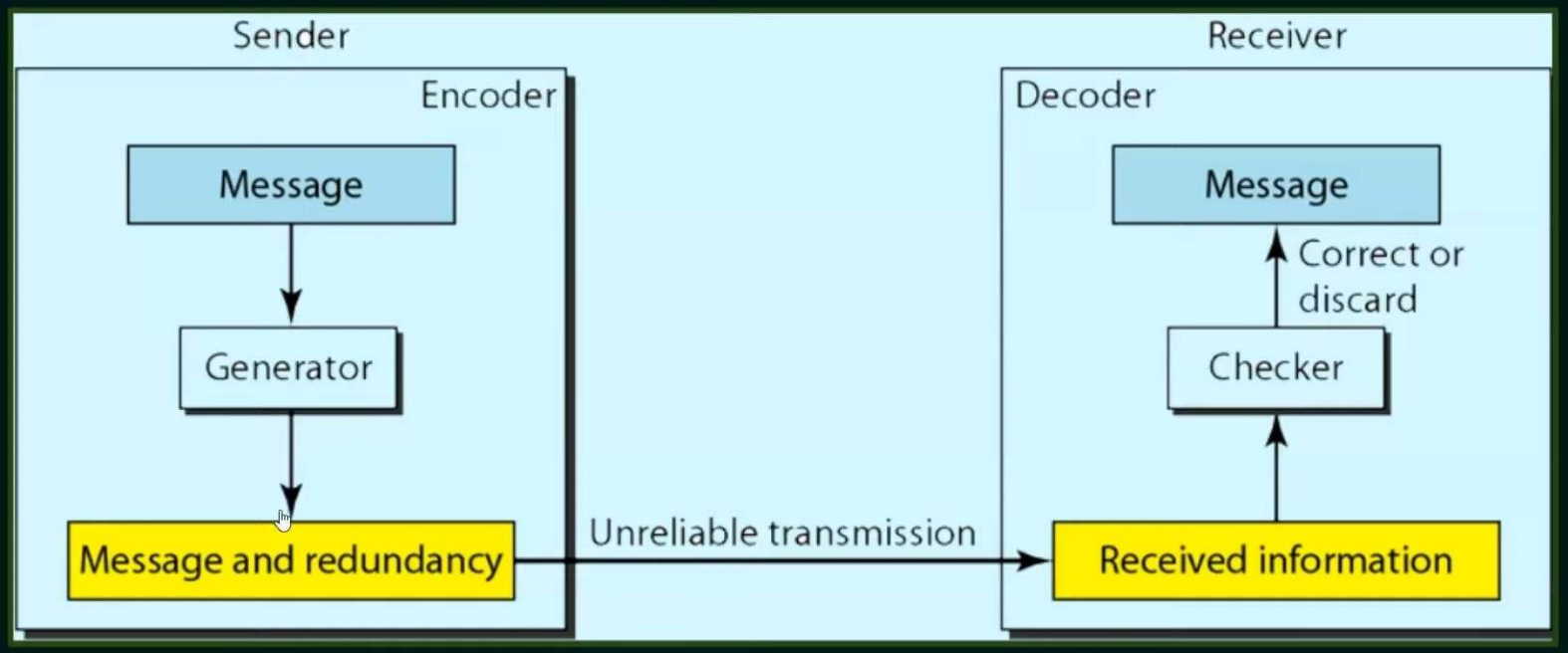
Error correction

It can be handled in two ways:

1) Receiver can have the sender retransmit the entire data unit.
2) The receiver can use an error-correcting code, which automatically corrects certain errors.

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ERROR DETECTION/CORRECTION



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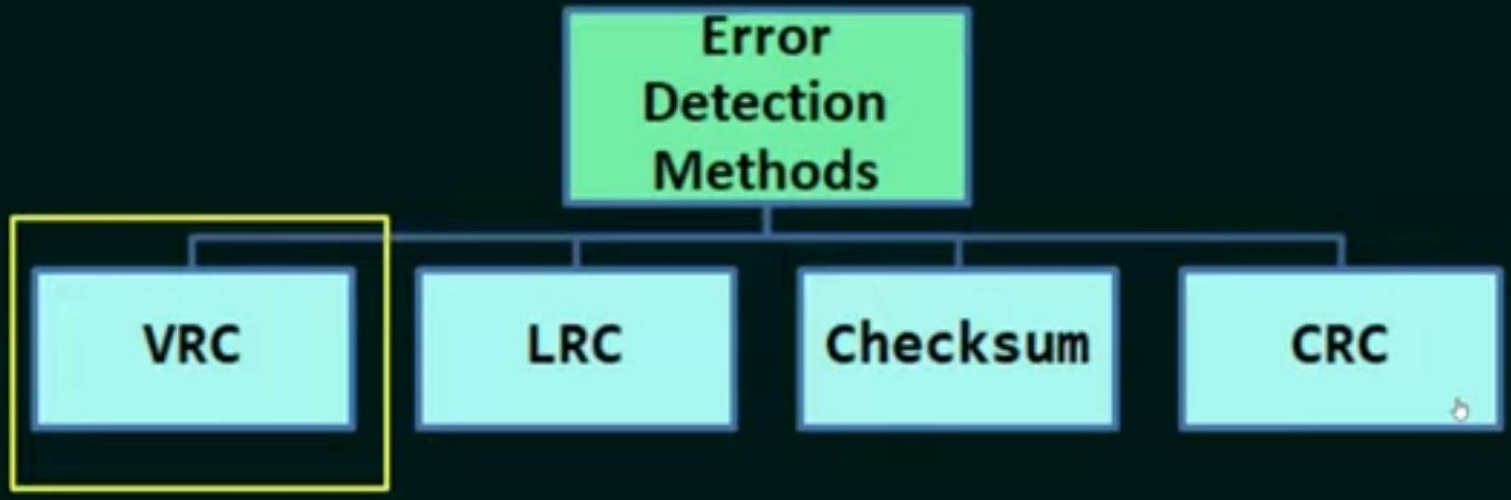
Error detection techniques

Four types of redundancy checks are used in data communications. They are:

1. Vertical Redundancy Check (VRC)
2. Longitudinal Redundancy Check (LRC)
3. Checksum
4. Cyclic Redundancy Check (CRC)

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Error detection

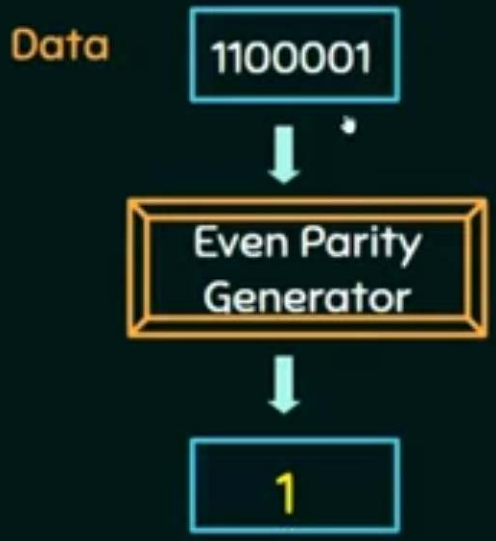


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VRC

- ★ Vertical Redundancy Check.
- ★ It is also called as parity check.

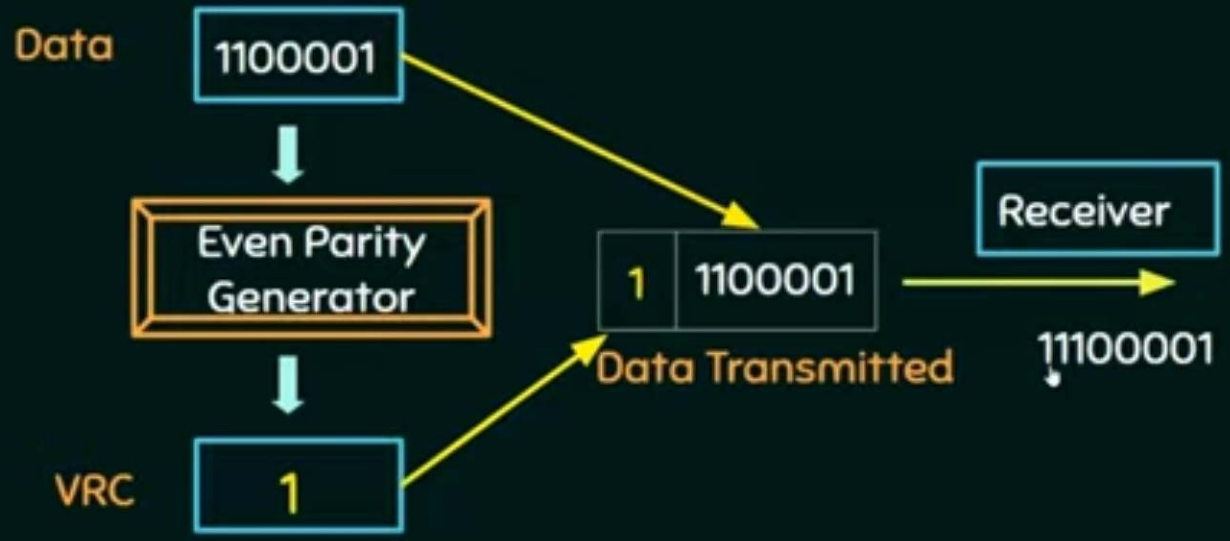
```
VRC: 1; if odd number of 1's
VRC: 0; if even number of 1's
```



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VRC

- ★ Vertical Redundancy check.
- ★ It is also called as parity check.



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Performance of VRC

- ★ It can detect single bit error.
- ★ It can detect burst error only if the number of errors is odd.

Sender: 11100001 -> Transmission Error 10100001 -> Receiver rejects this data.

Sender: 11100001 -> Transmission Error 10100101 -> Receiver accepts this data.

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LRC

- ★ Longitudinal Redundancy Check.
- ★ In LRC, a block of bits is organized in rows and columns.
- ★ a.k.a Two Dimensional parity.
- ★ The parity bit is calculated for each column and sent along with the data.
- ★ The block of parity acts as the redundant bits.

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LRC – Example

Find the LRC for the data blocks 11100111 11011101 00111001 10101001 and determine the data that is transmitted?

Odd no. of 1's	1
Even no. of 1's	0

1	1	1	0	0	1	1	1
1	1	0	1	1	1	0	1
0	0	1	1	1	0	0	1
1	0	1	0	1	0	0	1
1	0	1	0	1	0	1	0

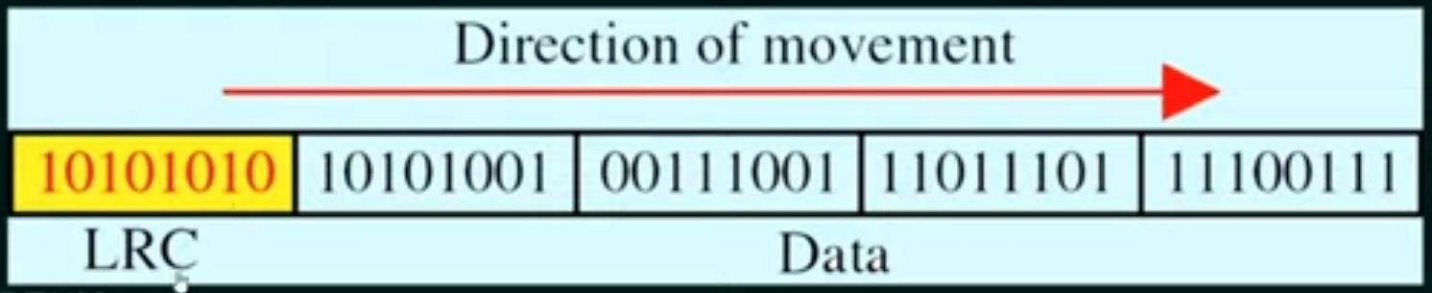
LRC →

∴ ← even

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LRC – Example

	1	1	1	0	0	1	1	1
	1	1	0	1	1	1	0	1
	0	0	1	1	1	0	0	1
	1	0	1	0	1	0	0	1
LRC →	1	0	1	0	1	0	1	0



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Performance of LRC

- ★ LCR increases the likelihood of detecting burst errors.
- ★ If two bits in one data units are damaged and two bits in exactly the same positions in another data unit are also damaged, the LRC checker will not detect an error.

	1	1	1	0	0	1	1	1
	1	1	0	1	1	1	0	1
	0	0	1	1	1	0	0	1
	1	0	1	0	1	0	0	1
LRC →	1	0	1	0	1	0	1	0

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Checksum

Checksum = Check + sum.

Sender side – Checksum Creation.

Receiver side – Checksum Validation.

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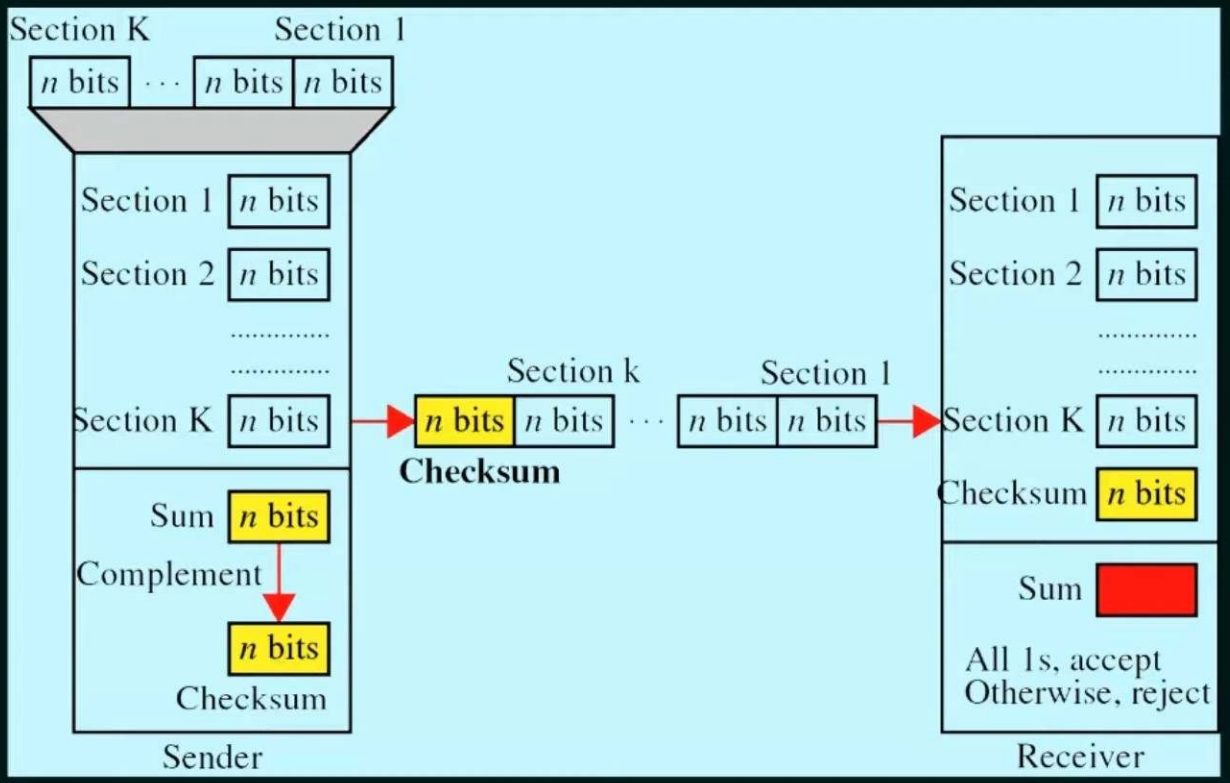


Checksum – Operation at sender side

1. Break the original message in to 'k' number of blocks with 'n' bits in each block.
2. Sum all the 'k' data blocks.
3. Add the carry to the sum, if any.
4. Do 1's complement to the sum = Checksum.

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Checksum



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Checksum – Example

Consider the data unit to be transmitted is:

10011001111000100010010010000100

10011001	11100010	00100100	10000100

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Checksum – Example

10011001	11100010	00100100	10000100

Carry	1	1	1	1	1			
	1	0	0	0	0	1	0	0
	0	0	1	0	0	1	0	0
	1	1	1	0	0	0	1	0
	1	0	0	1	1	0	0	1
1 0	0	0	1	0	0	0	1	1
							1	0



add it to the value

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Checksum – Example

10011001	11100010	00100100	10000100



Carry	1	1	1	1	1			
	1	0	0	0	0	1	0	0
	0	0	1	0	0	1	0	0
	1	1	1	0	0	0	1	0
	1	0	0	1	1	0	0	1
10	0	0	1	0	0	0	1	1
							1	0
	0	0	1	0	0	1	0	1
CHECKSUM	1	1	0	1	1	0	1	0

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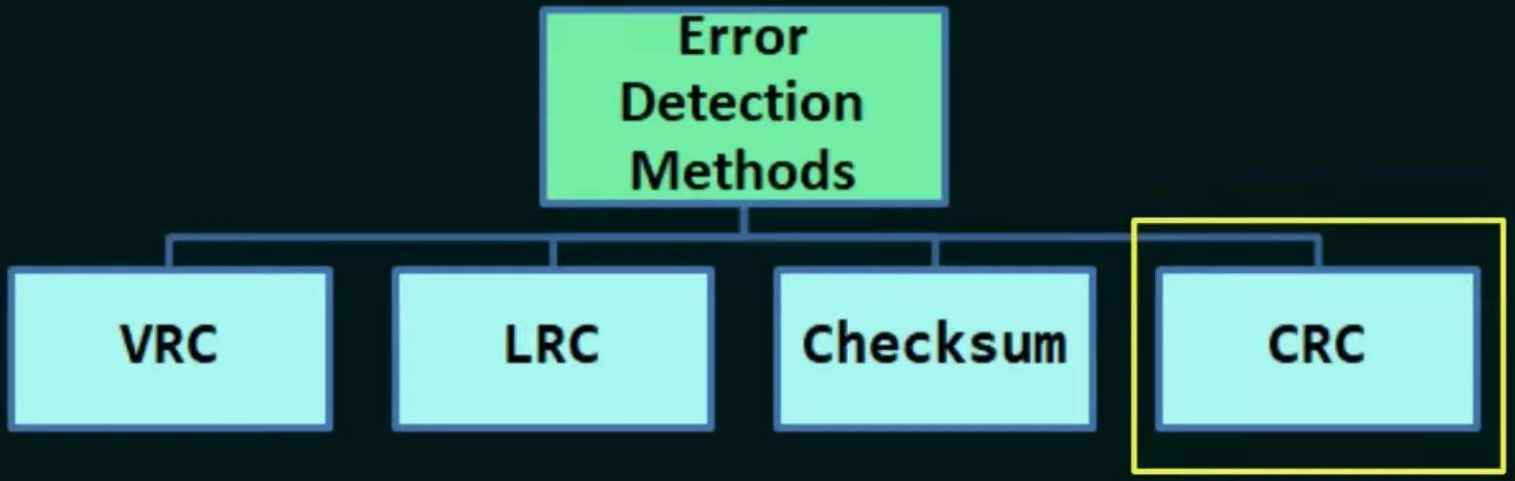
Checksum – Example

11011010	10011001	11100010	00100100	10000100	
Carry	1 1 1 1 1 1				
	1 0 0 0 0 1	0 0			
	0 0 1 0 0 1	0 0			
	1 1 1 0 0 0	1 0			
	1 0 0 1 1 0	0 1			
	1 1 0 1 1 0	1 0			check sum
10	1 1 1 1 1 1	0 1			
		1 0			
✓	1 1 1 1 1 1	1 1			



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Error detection



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CRC

Find the CRC for the data blocks 100100 with the divisor 1101?

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CRC Generation at sender side

1. Find the length of the divisor 'L'.
2. Append 'L-1' bits to the original message.
3. Perform binary division operation.
4. Remainder of the division = CRC.

Note:

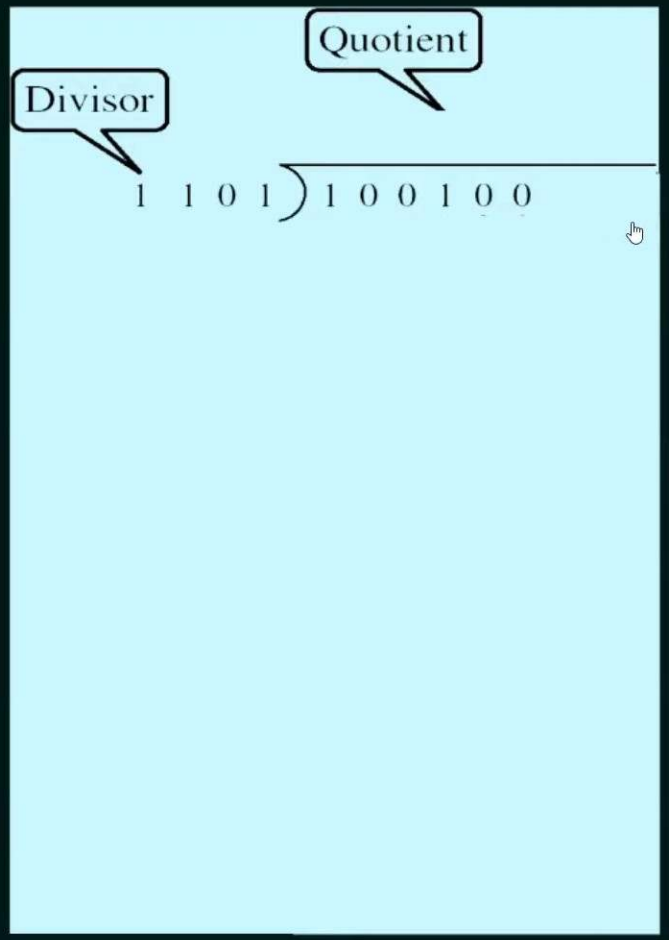
The CRC must be of L-1 bits.

A	B	A XOR B
0	0	0
0	1	1
1	0	1
1	1	0

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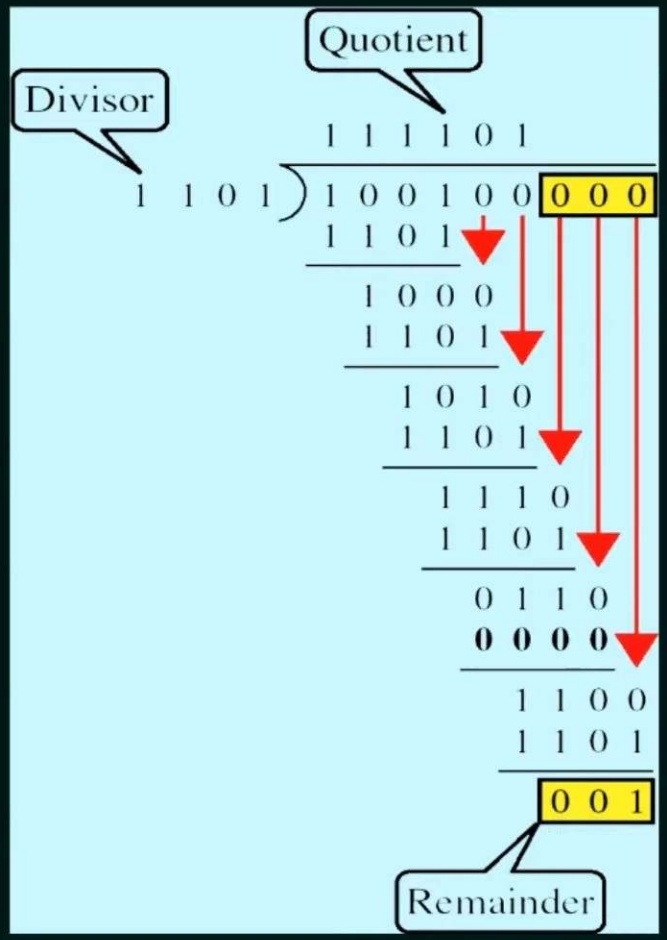
CRC

L= 4; So, 3 0's are appended to the message.



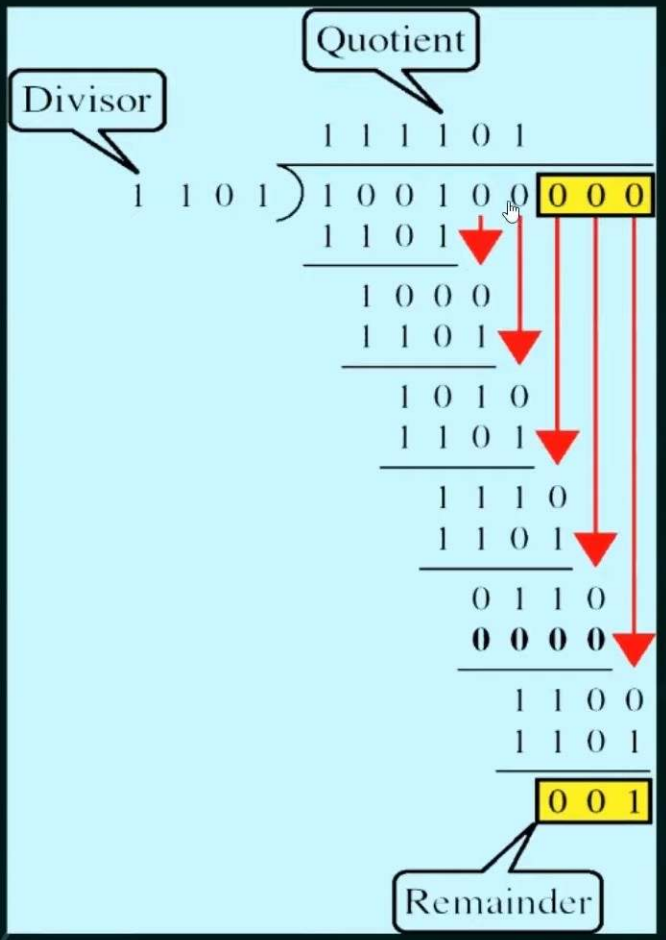
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CRC



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CRC

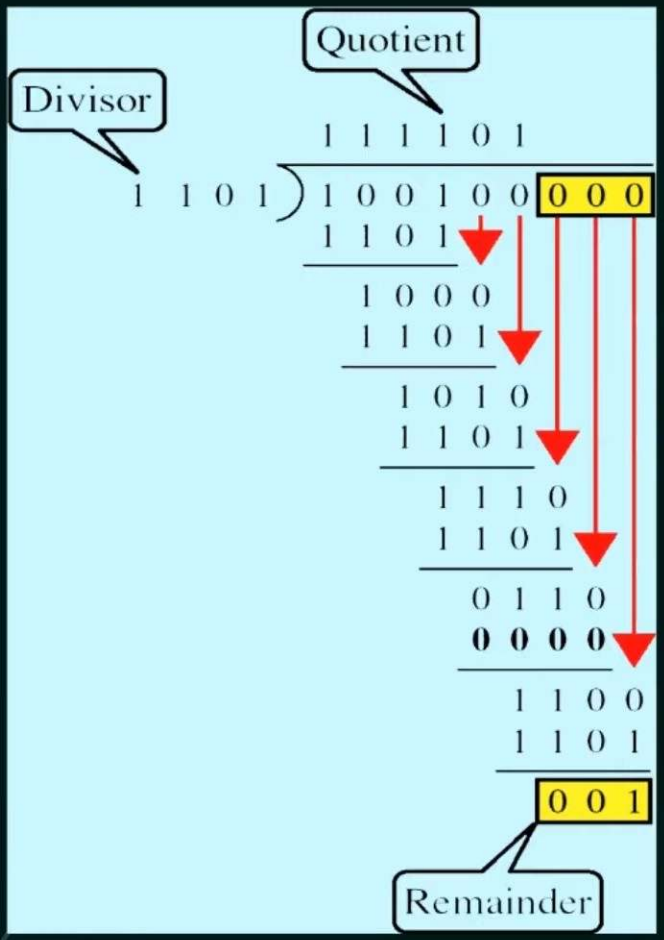


CRC: 001

Data Transmitted: 100100001

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CRC



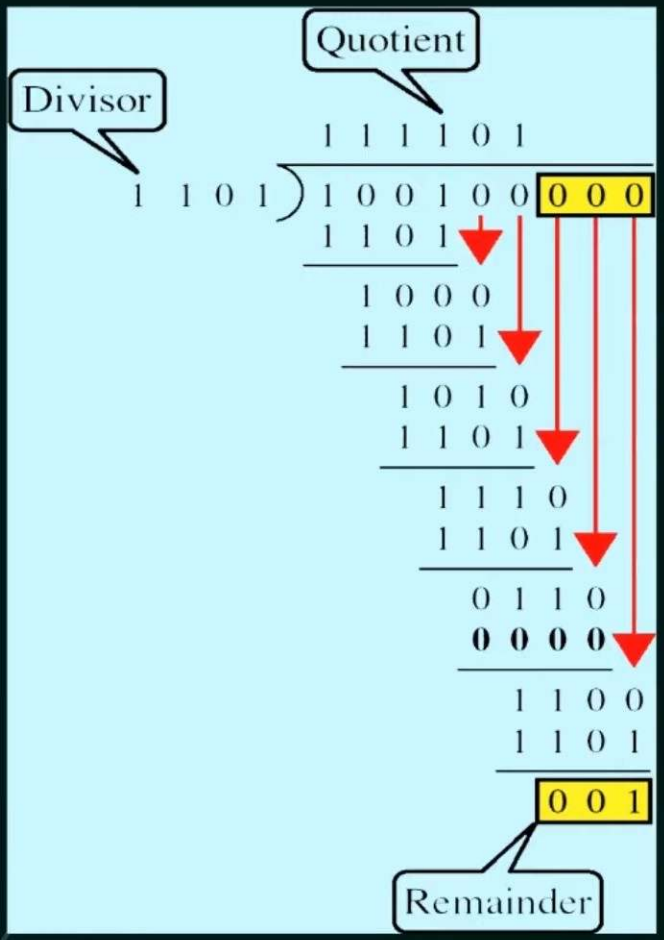
CRC: 001

Data Transmitted: 100100001

Data

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CRC



CRC: 001

Data Transmitted: 100100001

CRC

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CRC

Data Received by the receiver : 100100001

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CRC
1 1 1 1 0 1
1 1 0 1 | 1 0 0 1 0 0 0 0 1
1 1 0 1
1 0 0 0
1 1 0 1
1 0 1 0
1 1 0 1
1 1 1 0
1 1 0 1
0 1 1 0 1
1 1 0 1
0 0 0
Data Accepted
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